DC Charging (CCS2)-III

Introduction

The rapid adoption of electric vehicles (EVs) necessitates robust and intelligent charging infrastructure. While the physical connection of a charger to an EV might seem straightforward, the underlying communication protocols are complex, ensuring safety, efficiency, and interoperability. This lecture delves into the intricate communication sequences involved in DC charging using the CCS2 standard, building upon the foundational charging sequence discussed previously. We will explore how electric vehicle supply equipment (EVSE) and electric vehicles (EVs) establish a reliable digital dialogue, focusing on the critical role of Power Line Communication (PLC) and the sophisticated Signal Level Attenuation Characterization (SLAC) procedure. Understanding these protocols is paramount for developing and maintaining the next generation of smart charging solutions, addressing challenges like signal interference in multi-charger environments.

Core Concepts

Term	Definition	Significance
`CCS2`	Combined Charging System Type 2, a global standard for DC fast charging.	Defines the physical connector and communication protocols for high-power DC charging, ensuring interoperability.
`PLC`	Power Line Communication, a technology that uses electrical wiring to transmit data.	Enables high-level digital communication between EV and EVSE over the same CP and PE lines used for control signals.
`SLAC`	Signal Level Attenuation Characterization, a protocol to establish a logical connection between physically connected PLC modems.	Crucial for preventing crosstalk and ensuring that an EV communicates only with its physically connected EVSE in multi-charger environments.
`Crosstalk`	Unwanted signal interference between communication channels, often due to	A significant challenge in charging stations with multiple EVSEs, potentially leading to

Term	Definition	Significance
	shared wiring or close proximity.	incorrect logical connections between EVs and EVSEs.
`EVSE`	Electric Vehicle Supply Equipment, commonly known as a charging station.	The infrastructure component that supplies electrical energy to charge the EV, managing power delivery and communication.
`EVCC`	Electric Vehicle Communication Controller, the communication unit within the EV.	Manages the EV's side of the communication protocol, including initiating SLAC and requesting charging parameters.
`SECC`	Supply Equipment Communication Controller, the communication unit within the EVSE.	Manages the EVSE's side of the communication protocol, responding to EV requests and controlling power delivery.
`HomePlug Green PHY`	A specific standard for PLC used in EV charging.	Specifies the technical details for data transmission over power lines, including frequency bands and data rates (2-28 MHz, 4.5-10 Mbps).
`Attenuation`	The reduction in signal strength during transmission.	Measured by SLAC to determine the "best" communication path, indicating the most direct and physically connected EVSE.
`BCB Toggle`	A validation step in SLAC involving the rapid opening and closing of the S2 switch, changing the CP voltage.	Used to confirm the physical connection between an EV and EVSE by observing a low-frequency signal change, especially when attenuation results are ambiguous.

Detailed Analysis

1. Recap of DC CCS2 Charging Sequence

The DC CCS2 charging process follows a well-defined sequence, documented in standards like IEC 61851-23 and AIS 138. This sequence ensures a safe and controlled charging experience:
* **Mating**: The physical connection of the charger plug to the vehicle inlet.
* **Initialization**: The PLC modem establishes a logical connection, initiating high-level communication.
* Initially, the CP pin voltage is at 12V (State A).
* Upon mating, it transitions to State B1 (around 9V).
* The EVSE then activates PWM, moving to State B2 (still around 9V, but with PWM pulses, typically 5% duty cycle).
* High-level communication begins, and the system enters State C2 (around 6V).
* **Cable Check**: Continuous monitoring of insulation resistance between DC+ and DC- lines, ensuring it remains above 100 kOhms (typically observed in megaohms).
* **Pre-charging**: A controlled initial current flow (e.g., 1.95 A) to minimize inrush current due to voltage differences between the EVSE output and the battery pack. This prevents damage to components.
* **Charging**: The main power transfer phase, where the vehicle's Battery Management System (BMS) requests specific current and voltage levels, and the EVSE responds accordingly (e.g., 15 A).
* **Power Down**: As the battery reaches its desired State of Charge (SOC), the BMS signals the EVSE to gradually reduce power.
* **Unlock and Unmate**: The charging process concludes, the connector is unlocked, and the plug can be safely removed.

The CP pin voltage levels and duty cycles are critical indicators of the charging state. For instance, a 100% duty cycle on the CP line signifies State A, while a 3-7% duty cycle (e.g., 5%) indicates State B2, where high-level communication via PLC is active. The PP pin also plays a role, with its resistance (e.g., 1.385 kOhm, within a range of 891-1588 Ohm) indicating a valid connection and immobilizing the vehicle.

2. PLC-based Communication and Architecture

High-level communication in CCS2 DC charging relies on Power Line Communication (PLC), specifically using the HomePlug Green PHY (HPGP) standard.
* **Standardization**: IEC 61851-23 mandates PLC for high-level communication, with HPGP being a subset of the HomePlug AV standard.
* **Data Transmission**: HPGP operates in a wideband frequency range of 2 to 28 MHz, achieving digital data rates between 4.5 to 10 Mbps. This high-frequency

signal is superimposed on the 1 kHz PWM signal on the CP and PE lines.
* **System Architecture**:
* **EVSE Side**: An AC supply powers the EVSE, which includes an AC-to-DC converter and an isolated DC-to-DC Power Conversion Unit (PCU). The Supply Equipment Communication Controller (SECC) manages communication, powered by a small AC-to-DC converter from the same AC supply. A PLC modem connects the SECC to the CP and PE lines.
* **EV Side**: The Electric Vehicle Communication Controller (EVCC) communicates via its own PLC modem to the CP and PE lines.
* **Challenge of Crosstalk**: In public charging stations with multiple EVSEs connected to the same AC supply, the high-frequency PLC signals can "crosstalk" between different charging points. This means a signal from EV1 intended for EVSE1 might inadvertently reach EVSE2's PLC modem, leading to potential logical connection errors. Crosstalk can occur due to:
* High-frequency signals navigating through shared copper wires.
* Radiated coupling to DC+ and DC- lines, then through isolation transformers in the PCU to the supply line.
* Close proximity of control wires, especially in EVSEs with multiple charging guns.
* Shared ground loops, poor EMI design, and unfiltered power supplies.

3. The Signal Level Attenuation Characterization (SLAC) Procedure

To mitigate crosstalk and ensure that an EV communicates only with its physically connected EVSE, the Signal Level Attenuation Characterization (SLAC) procedure is performed. SLAC is a protocol that measures signal attenuation between PLC modems to establish the correct logical connection. It is defined in the HomePlug Green PHY v1.1.1 specification and detailed in ISO 15118-3 and DIN SPEC 70121 standards. SLAC is initiated after the charging state reaches B2.
* **1. Parameter Exchange/Setup**:
* The EV initiates SLAC by sending a broadcast message, `CM_SLAC_PARM.REQ`, to all EVSEs in the charging station.
* All listening EVSEs respond with a `CM_SLAC_PARM.CNF` confirmation message. This message includes parameters like the number of "sounding" (attenuation indication) signals the EV should send.
* This step sets up SLAC parameters such as RunID, allowed frequency range, timings, and attenuation thresholds.
* **2. Discovery of PLC Nodes**:
* The EV broadcasts `CM_START_ATTEN_CHAR.IND` three times, acting as a "wake-up call" for EVSEs that sent confirmation messages.
* The EV then sends a preferred amount (typically 10) of "sounding messages." Each sounding message comprises 58 different frequency groups, corresponding to

narrow bands within the 2-28 MHz PLC frequency range.
* For each `CM_MNBC_SOUND.IND` message received, the EVSE's PLC modem measures the attenuation for all 58 frequency groups and generates a `CM_PROFILE.IND` message.
* After receiving all 10 sounding messages, the EVSE's PLC modem sends a `CM_ATTN_CHAR.IND` message containing the average attenuation values for the 58 frequency groups.
* The EV acknowledges these attenuation profiles and begins analysis for a matching decision.
* **3. Matching Decision**:
* The EV's PLC modem analyzes the received attenuation profiles. If multiple EVSEs responded (due to crosstalk), the EV decides which EVSE to logically connect to.
* The EVSE with the least attenuation value is chosen for the logical connection.
* If the results are clear (e.g., only one EVSE responded, or one has significantly lower attenuation), the EV proceeds by sending `CM_ATTEN_CHAR_RES` to the chosen EVSE.
* If the results are ambiguous or "undecided" (e.g., multiple EVSEs show similar low attenuation), a validation process is required.
* **4. Validation (Optional)**:
* The EV sends a `CM_VALID_CPLT.REQ` message to start validation, which the EVSE confirms with `CM_VALID_CPLT.CNF`.
* The EV then performs a "BCB toggle" by rapidly closing and re-opening its S2 switch. This causes the CP pin voltage to toggle between B2 (9V) and C2 (6V) states multiple times (e.g., two BCB toggles). This is a low-frequency signal change that is only observable by the physically connected EVSE.
* The EV sends `CM_VALID_CPLT.REQ` again.
* The physically connected EVSE confirms the message, including the number of detected BCB toggles. This confirms the physical connection, allowing the EV to definitively choose the correct EVSE.
* **5. Key Exchange**:
* Once the correct EVSE is chosen, the EV informs it by sending a `CM_SLAC_MATCH.REQ` message.
* The EVSE confirms with `CM_SLAC_MATCH.CNF`, which contains the logical Network Membership Key (NMK) and Network ID (NID).
* The EV sets these received keys in its PLC modem using a `CM_SET_KEY.REQ` message, which is confirmed by the PLC modem.
* The PLC modem automatically reboots. After rebooting, the PLC modems of the EV and EVSE are logically connected, ready for high-level communication.

4. Stateless Address Autoconfiguration (SLAAC)

Following the successful SLAC procedure and logical connection, Stateless Address Autoconfiguration (SLAAC) takes place.
* SLAAC is an IPv6 mechanism that assigns local-link IP addresses to the EV and EVSE PLC modems.
* This step ensures that once logically connected, the EV and EVSE have unique network addresses, enabling seamless data transmission and further high-level communication over the network layer (Layer 3 of the OSI model).

Key Takeaways

- [ ] The DC CCS2 charging sequence involves distinct stages: mating, initialization, cable check, pre-charging, charging, power down, and un-mating.
- [ ] PLC-based communication, specifically HomePlug Green PHY, is essential for high-level data exchange between EV and EVSE over the CP and PE lines.
- [ ] Crosstalk is a significant challenge in multi-EVSE charging environments, where high-frequency PLC signals can interfere, leading to incorrect logical connections.
- [ ] The SLAC (Signal Level Attenuation Characterization) procedure is designed to ensure that an EV's PLC modem logically connects only with its physically connected EVSE.
- [ ] SLAC involves several steps: parameter exchange, discovery of PLC nodes (using sounding messages and attenuation profiles), a matching decision based on least attenuation, and an optional validation step (BCB toggle).
- [ ] The BCB toggle, a low-frequency signal change, is a crucial validation step to confirm physical connection when attenuation measurements are ambiguous.
- [ ] After SLAC, key exchange establishes a secure logical connection, followed by a PLC modem reboot.
- [ ] SLAAC (Stateless Address Autoconfiguration) assigns IPv6 local-link addresses to the logically connected EV and EVSE PLC modems, enabling network-layer communication.

Conclusion

The detailed examination of the DC CCS2 charging sequence and the SLAC procedure highlights the sophisticated engineering required to ensure reliable and safe electric vehicle charging. From the initial physical connection to the

establishment of a robust logical communication link, every step is meticulously designed to prevent errors and optimize performance. The SLAC protocol, in particular, addresses the critical challenge of crosstalk in complex charging environments, guaranteeing that the right EV communicates with the right EVSE. This foundational communication layer, operating at the data link layer of the OSI model, paves the way for higher-level communication protocols and advanced smart charging functionalities, which will be explored in subsequent discussions. Understanding these intricate processes is vital for anyone involved in the design, deployment, or maintenance of modern EV charging infrastructure.